

Anglais – Synthèse Centrale-Supélec

Durée de l'épreuve : 4h

Rédiger en anglais et en 500 mots (plus ou moins 10%) une synthèse des documents proposés, qui devra obligatoirement comporter un titre. Indiquer avec précision, à la fin du travail, le nombre de mots utilisés (titre inclus).

Ce sujet comporte les quatre documents suivants :

- une photographie de Frans Krajcberg, tiré de son reportage photo « Queimadas » (1985) ;
- un article publié dans *The Guardian* le 12 octobre 2021 ;
- un article éducatif publié dans *National Geographic* le 15 janvier 2020 ;
- un article publié dans *science.org* le 19 décembre 2017.

L'ordre dans lequel se présentent les documents est arbitraire et ne revêt aucune signification.

Frans Krajcberg, "Queimadas", 1985



Frans Krajcberg (1921- -) in 1985 while shooting "Queimadas", a long photographic report on deforestation in Mato Grosso (Brazil). "I am a burnt man. Fire is death, the abyss. Fire has always been with me. My message is tragic. I show crime. I bring the documents, I collect them and add more, I want to give my revolt the most dramatic and violent face. I want my work to be a burn reflex"

More than 100 moorland fires reported in England in past four days

The Guardian | Robyn Vinter | Tue 12 Oct 2021

More than 100 moorland fires have been reported across England's national parks in the past four days, a fivefold increase on last year. Just weeks before countries gather in Glasgow for Cop26, the UN climate summit, thousands of acres of moorland across the north of England have been torched as part of annual burning carried out by the grouse shooting industry, a practice wildlife campaigners describe as "absurd" and "illogical on so many grounds".

Despite legislation brought in earlier this year that banned burning on some of the country's most valuable peatland, figures compiled by the wildlife campaign group Wild Moors, and shared with Unearthed, the investigative arm of Greenpeace, and the *Guardian* show the number of fires has risen dramatically.

This is part of a growing trend for burning peat moorland, mostly in the North York Moors, the Peak District and the Yorkshire Dales, which causes ecological damage, releases large amounts of carbon dioxide into the atmosphere and can lead to flooding. During burning season, when burning is permitted between 1 October and 15 April, gamekeepers use blowtorches to strip back heather, allowing new shoots to grow, which provide food for grouse.

While most of the fires are legal, Natural England told the *Guardian* it was investigating two locations where illegal burning may have been carried out. Volunteers recorded 37 burns in October 2020. There were 719 for the full season last year, a rise from 550 in 2019.

Luke Steele, the director of Wild Moors, said: "Whether it's legal or illegal, it's outright wrong, because these are our biggest natural carbon stores and we're literally setting them on fire for grouse shooting." [...]

Peatland is the UK's largest natural CO₂ store, locking in an estimated 3.2bn tonnes of CO₂, as well as providing nesting and feeding grounds for many wading birds and important habitats for rare insects and plants.

The Heather and Grass Burning (England) Regulations 2021, which came into force in May this year, ban burning on land in conservation and special protection areas where there is peat more than 40cm in depth. This leaves about 40% of the UK's peatland vulnerable to burning, falling short of the recommendations of the government's climate crisis committee, which called for a total ban on burning on peatland as part of a wider package of changes to the way land is managed, in order to reach net zero emissions by 2050. Kate Blagojevic, Greenpeace UK's head of climate, said: "It's obvious that the government's regulations are worse than toothless and completely failed to stop this absurd practice that damages both the climate and wildlife.

"A comprehensive ban should be introduced immediately along with concrete measures to fully or highly protect at least 30% of our land and seas by 2030. Anything less would be a major embarrassment for the UK government. There are better ways to welcome world leaders to a crucial climate summit than the sight of smoke and flames engulfing our largest carbon store."

The Ecological Benefits of Fire

National Geographic | Wednesday, January 15, 2020

The year 2019 marked 75 years of Smokey Bear, the advertising icon of the United States Forest Service who encourages visitors and campers to prevent wildfires. Wildfires are destructive forces that can result from natural causes (like lightning), human-caused accidents (like cigarettes and campfires), or deliberate acts of arson. Despite Smokey's education campaigns, wildfires burned about four million hectares (10 million acres) of land during 2017, and in 2018, a single California wildfire, the "Camp Fire" destroyed nearly 20,000 structures and killed more than 80 people, with

insured losses topping \$10 billion. However, while these frightening and negative consequences dominate news headlines, forest fires have a positive side. Controlled use of wildland fires for positive environmental effects is common around the world.

While a wildfire refers to an unintentional, uncontrolled fire, the term "wildland fire" is broader and includes fires purposefully set as part of prescribed burns. While all fires have the potential to become dangerous to property and life, prescribed, or controlled, burns are planned extensively and performed with tight safety parameters. Humans have been performing such burns for thousands of years and for multiple reasons, but, today, they are mainly used to promote ecological health and prevent larger, more damaging, uncontrolled fires.

It might seem counterintuitive that a fire, which burns plant life and endangers animals within an ecosystem, could promote ecological health. But fire is a natural phenomenon, and nature has evolved with its presence. Many ecosystems benefit from periodic fires, because they clear out dead organic material—and some plant and animal populations require the benefits fire brings to survive and reproduce.

For example, as dead or decaying plants begin to build up on the ground, they may prevent organisms within the soil from accessing nutrients or block animals on the land from accessing the soil. This coating of dead organic matter can also choke outgrowth of smaller or new plants. When humans perform a prescribed burn, the goal is to remove that layer of decay in a controlled manner, allowing the other, healthy parts of the ecosystem to thrive. Moreover, nutrients released from the burned material, which includes dead plants and animals, return more quickly into the soil than if they had slowly decayed over time. In this way, fire increases soil fertility—a benefit that has been exploited by farmers for centuries.

Several plants actually require fire to move along their life cycles. For example, seeds from many pine tree species are enclosed in pine cones that are covered in pitch, which must be melted by fire for the seeds to be released. Other trees, plants, and flowers, like certain types of lilies, also require fire for seed germination.

Perhaps surprisingly, the animal casualties from wildfires are low—animals survive by burrowing into the ground or fleeing to safer areas. Conversely, fires can help rid an ecosystem of invasive species that have not adapted to regular wildland fires. While animals and plants within fire-prone ecosystems have adapted to thrive within a cycle of wildfires, invasive plants and animals are less likely to recover and could thus be controlled or even eradicated from the ecosystem they invaded.

Moreover, prescribed burns are well established as a way to prevent more devastating naturally occurring fires. The buildup of decaying organic matter on the ground is fuel for wildfires. Without periodic fire to clear this out, a naturally occurring fire may grow and move quickly, doing much more damage than a prescribed burn—and without its safety parameters.

In the end, it is true that the burden of preventing uncontrolled wildfires lies with humanity. Smokey Bear's message is right—nearly 85 percent of wildfires originate from human activity, and we have to take action to prevent these damaging fires. But suppression is not enough. Nature needs fire, and ecologically benefits from periodic burning. In fact, suppression alone might make matters worse, depriving nature of its equivalent of spring cleaning and leading to hotter, larger blazes when built-up forest decay finally catches flame. Understanding and appreciating the benefits of fire is the only way to truly keep our homes, population, and ecosystem safe from its dangers.

Ecosystems could once bounce back from wildfires. Now, they're being wiped out for good

Study suggest that climate change is making devastation permanent
Science.org | 19 Dec 2017 | Lakshmi Supriya

Uncontrollable infernos that have torched about half a million hectares and displaced more than 100,000 people have made this the worst wildfire year yet for California. From such ashes, ecosystems usually bounce back, but a new study reveals this is no longer a guarantee. Thanks to climate change, areas ravaged by wildfires may never recover, wiping out entire ecological communities forever.

Wildfires are a natural part of many environments. [...] But fires are only good if they serve their specific purpose. If they burn too long, or the ground stays dry too long, ecosystems can't recover. Given that climate change can lead to more fires and longer droughts, researchers have wondered how forests are coping, and whether they are getting a good start on their way to becoming a new generation of trees.

To find out, ecologist Camille Stevens-Rumann of Colorado State University in Fort Collins and colleagues studied about 1500 sites in the conifer forests of the U.S. Rocky Mountains that had seen 52 wildfires between 1988 and 2011. The areas spanned elevations from about 700 to 2800 meters above sea level and comprised various types of dry and wet pine forests. The researchers collected seeds from the sites between 2010 and 2014, and, along with previously published data, analyzed the plots' seedling presence and density. By comparing these data to the seedlings in sites that had not burned, they determined the ability of forests to regenerate. In addition, they combined these data with climate—temperature and moisture—information to see how it affected tree regrowth.

They found a dramatic difference in tree regeneration after fires late last century compared with fires earlier this century. The proportion of sites with no regrowth almost doubled after 2000, from 19% to 32%, coinciding with increasing temperatures and more droughts, the team reports this month in *Ecology Letters*. Although forests that burned before 2000 have had more time to grow, the presence of seedlings in the first 5 years gives a good indication of future growth.

"Essentially either you have abundant seedlings and continue to grow, or you have none, and rarely is a site filled in," Stevens-Rumann says. Forests in the hottest and driest regions were most susceptible. So fires in these areas may cause landscapes and ecosystems to change, for example, from forests to grasslands or shrubs. And this may become a vicious cycle. Fires pump more carbon into the atmosphere, exacerbating climate change, and fires decimate the trees that would normally suck this carbon out of the sky.

Some forests in the moister, higher elevations may not grow the old trees back, but may change to different tree species that are better suited to the hotter, drier weather. This might be a cue for future forest management, Stevens-Rumann says. Managers may want to plant species that are adapted to the current and future climate, not the climate of the past, she says.

But, not all forests are in danger of dying out completely. As Stevens-Rumann acknowledges, the time span she and colleagues studied, 23 years, is very short compared with the life span of a forest. "Even among our sites, some of the forests we studied are regenerating very well," she says. Although some forests may grow back, she says, at the very least we can say that it is going to take much longer that it once did.